disassembler: planetary-scale sequence extraction from Logan

LOGAN-DISASSEMBLER

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`github.com/nekrut/disassembler`

A graph path-walker on Logan unitigs and a streaming assembler on Logan contigs, over the same petabase.

One gene, many planetary samples

Logan indexes the SRA petabase as **unitig graphs** and **assembled contigs**. Given one reference CDS, `disassembler` reconstructs full-length haplotypes from either, ending in the same codon MSA, haplotype catalog, and tree.

LOGAN UNIVERSE
4.99 M
subject genomes in the LexicMap index, the universe each query scans.

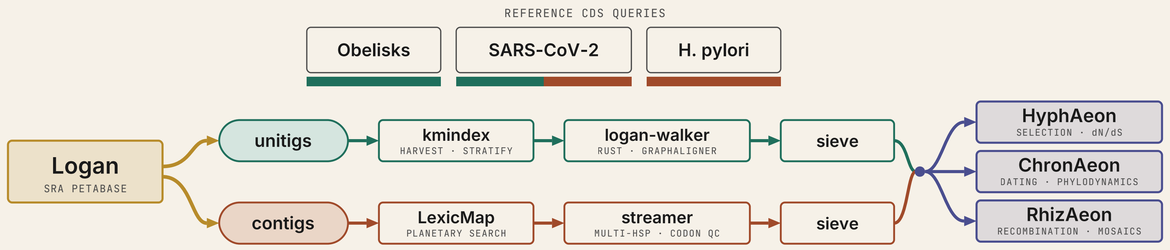
WALKER CONTAMINATION FILTER
0.555% → 0%
mislabeled spike accessions, raw unitig hits → CLEAN_FULL_LENGTH.

Two routes, one cohort shape

ROUTE A · UNITIG GRAPH

When the walker route wins

- **Shattered genes.** Zika NS5: ~50 contig fragments, one clean walk.
- **Divergent tails** a single-reference search never reaches.
- **Intra-host variation** kept, not collapsed: per-site allele frequencies.



ROUTE B · CONTIG STREAMER

When the LexicMap route wins

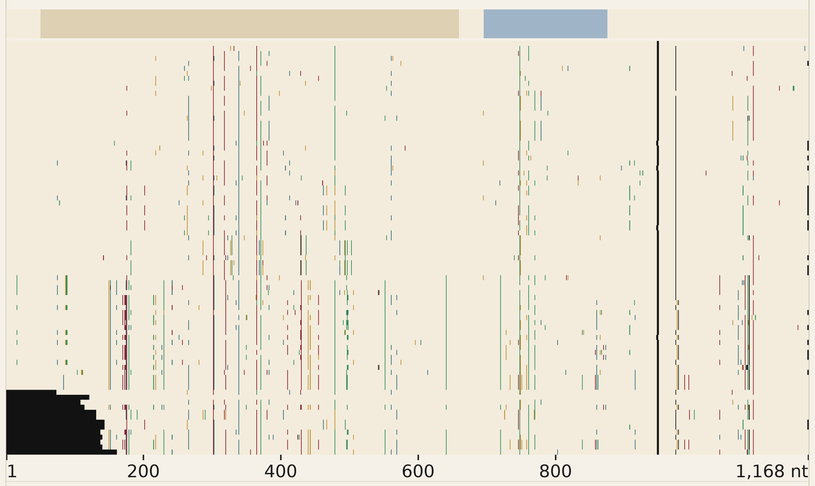
- **Split genes.** Collinear tiling stitches a 55/45 HSP pair that fails `qcovHSP ≥ 80`.
- **Planetary scale.** 24 GB streamed; memory is O(haplotypes), not O(rows).
- **Hands off to the walker** below 85% median identity.

ROUTE A · KMINDEX → WALKER

Obelisks across human + bacterial microbiomes

- **METATRANSCRIPTOMIC_BCT dominates**: obelisks live in bacterial microbiome transcriptomes.

KMINDEX CANDIDATES	MAX CONTAINMENT	SHARDS QUERIED	CENTROIDS
924 unique runs	0.97 alpha	141 meta + metaT	5 α β γ ε Ss



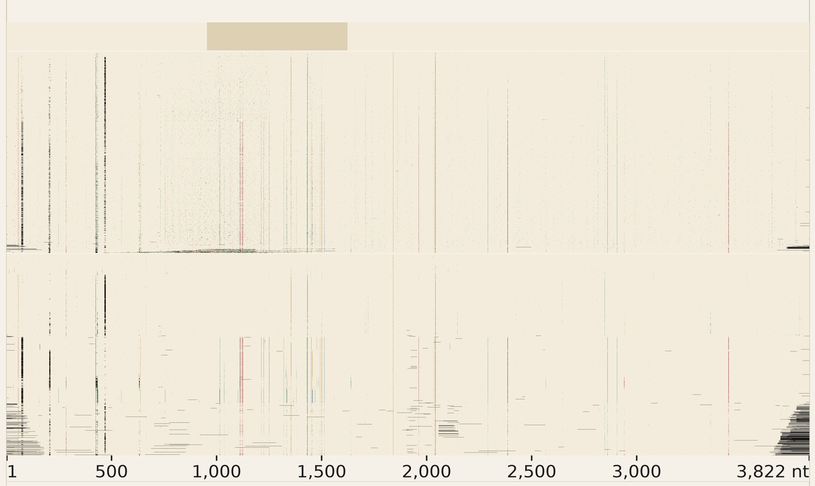
Obelisk-alpha. Reference strip on top: **Oblin-1** brown, **Oblin-2** blue. Rows sorted by edit count.
match A C G T gap

ROUTE A ROUTE B

SARS-CoV-2 spike: two routes, one reference

- **The routes are complementary.** Only the walker touches the divergent tail; only LexicMap-permissive catches the 1.15 M Alpha/Delta-era accessions strict loses.

LEXICMAP RAW HSPS	WALKER20K SUCCESS	CLEAN_FULL_LENGTH	PERM. SIEVED
5.88 M 24 GB	47.2% 9,448 / 20k	6,827 pid 98.75	79,621 of 1.15 M



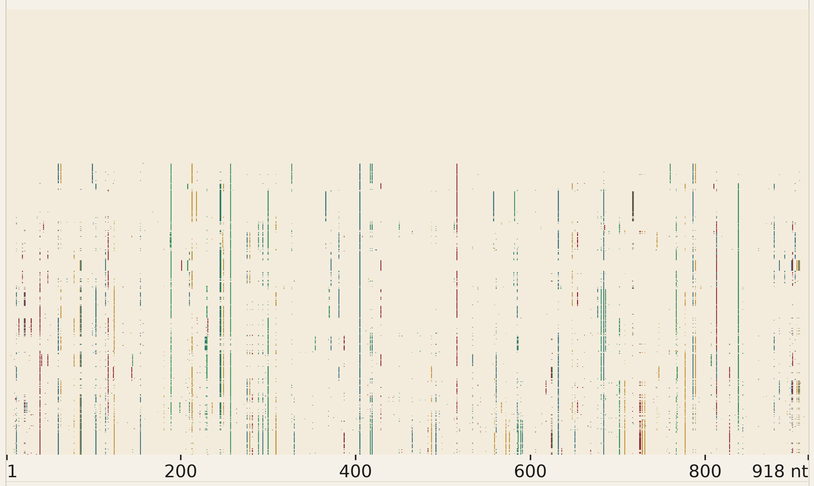
Shared spike reference on top, **RBD** brown. **Top:** kmindex → walker 20k. **Bottom:** LexicMap permissive. Columns align.
match A C G T gap

ROUTE B · LEXICMAP → STREAMER

Helicobacter pylori: 25-gene marker panel

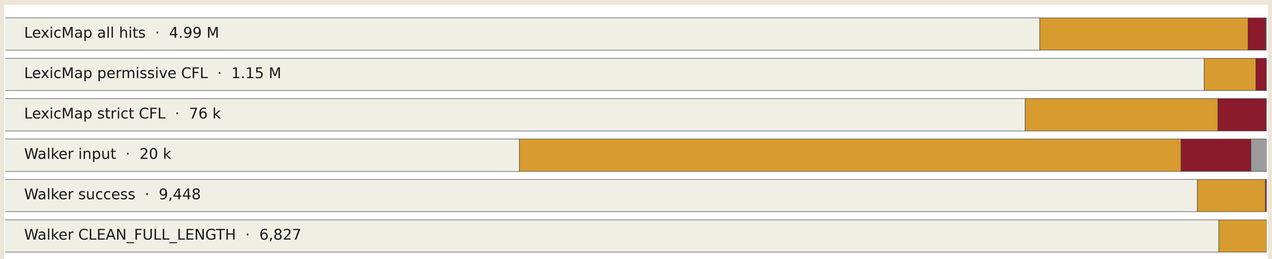
- **Four audit tiers per marker**, one reference frame: clean full-length through low-coverage fragment, every accession accounted for.

STRICT MARKERS	LOOSE MARKERS	AUDIT TIERS	HYPHAEON PASS
20	5 selection	4 per marker	oipA done



oipA strict panel: 1,042 clean accessions, sorted by substitution count.
match A C G T gap

SRA metadata: the walker drives contamination to zero



Six cohorts against the 2026-09-11 ENA snapshot, zoomed to 90–100%.
A labeled SARS-CoV-2 B mixed-sample plausible C mislabeled D no ENA record

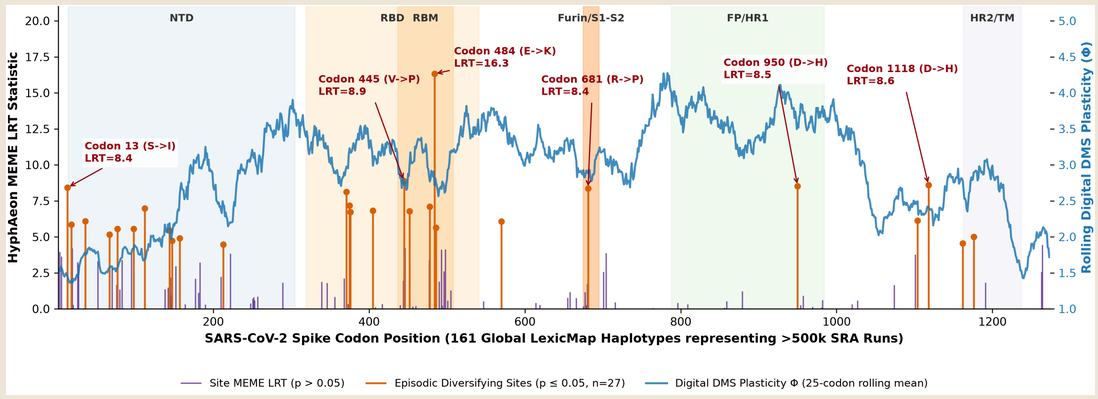
WALKER20K INPUT
0.555%
C-mislabeled, raw unitig hits

WALKER20K SUCCESS
0.011%
1 in ~9,500

CLEAN_FULL_LENGTH
0.000%
0 / 6,827

LEXICMAP PERMISSIVE
0.084%
no walker filter

HyphAeon: selection on the recovered cohort



27 diversifying sites across 161 global spike haplotypes, led by the antibody-escape hub **E484K**. Shaded bands are spike domains.

HAPLOTYPES	TOP SITE	RUNTIME
161 >500k SRA runs	E484K LRT 16.3	29.5 s no input tree

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`github.com/nekrut/disassembler` `zenodo 10.5281/zenodo.23737199` `github.com/veg/HyphAeon`